

Solution Architecture

Project: Employee Software Installation Request (ServiceNow)

1. Architecture Overview

The proposed solution is developed entirely on the **ServiceNow platform** using a **Custom Scoped Application**. The architecture follows a **layered and data-driven approach**, where employees submit software installation requests through the **Service Catalog**, and the requests are processed automatically using ServiceNow's native workflow components.

The solution utilizes **Flow Designer** for workflow automation, the **Approval Engine** for manager approvals, **Business Rules** for server-side automation, and standard ServiceNow tables for storing and tracking requests. This architecture ensures scalability, maintainability, and seamless integration with ServiceNow's built-in capabilities.

2. Layered Architecture

The solution is divided into three logical layers, each responsible for a specific set of functions.

Presentation Layer

This layer provides the user interface through which different users interact with the system.

Components

- **Service Catalog** – Allows employees to submit software installation requests.
- **My Requests** – Enables employees to track the status of their submitted requests.
- **My Approvals** – Allows managers to review, approve, or reject software requests.
- **Notifications** – Displays email and system notifications related to request status.

Application / Process Layer

This layer contains the business logic and workflow automation responsible for processing requests.

Components

- **Flow Designer** – Automates request routing, approval workflow, task creation, and notifications.
- **Approval Engine** – Manages approval records and approval decisions.
- **Business Rules** – Executes server-side logic such as automatic incident creation when software licenses are unavailable.
- **Catalog Tasks** – Generates and assigns fulfillment tasks to the Network/Software Support Team.

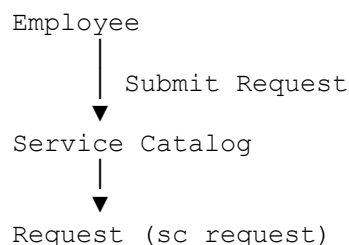
Data Layer

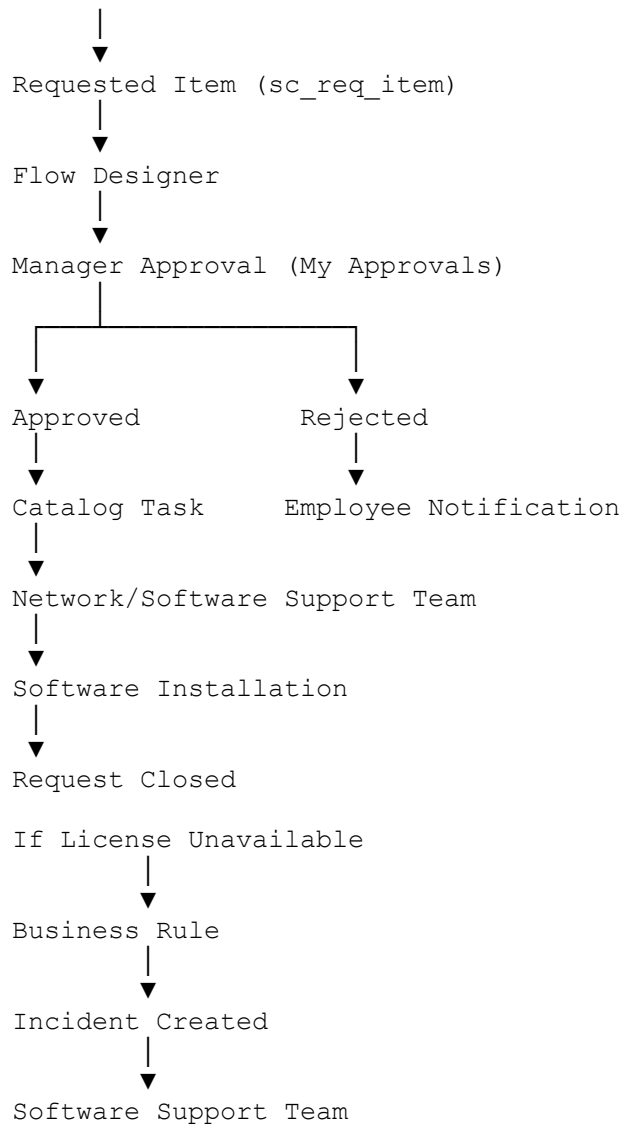
This layer stores all request information and workflow records using standard ServiceNow database tables.

Table	Purpose
sc_cat_item	Stores the Software Installation Service Catalog item.
sc_request	Stores the parent request submitted by the employee.
sc_req_item	Stores individual requested items and tracks their lifecycle.
sysapproval_approver	Stores manager approval records.
sc_task	Stores fulfillment tasks assigned to the IT/Network Support Team.
incident	Stores incidents created for software license shortages.
sys_user	Stores employee and manager information used for approval routing.

3. Component Interaction

The following sequence illustrates how the major components interact during the software installation request process.





4. Key Design Decisions

Several design decisions were made to ensure that the solution is reliable, maintainable, and aligned with ServiceNow best practices.

- The application is developed as a **Custom Scoped Application**, allowing all custom configurations, scripts, and workflows to remain isolated from the global scope.
- The solution uses the native **Service Catalog** instead of custom request forms, providing a standardized and user-friendly interface.
- **Flow Designer** is used to automate approvals, task creation, and notifications because it offers a modern, low-code approach that simplifies maintenance and future enhancements.

- **Business Rules** are reserved for simple server-side automation, such as creating incidents when software licenses are unavailable, ensuring a clear separation between workflow orchestration and backend logic.
- Standard ServiceNow tables are used instead of creating custom tables, improving compatibility with existing platform features.

5. Security Architecture

The solution implements role-based security to ensure that users can access only the information relevant to their responsibilities.

Role	Access Permissions
Employee	Can submit software requests and view only their own requests through My Requests .
Manager	Can review, approve, or reject requests assigned to them through My Approvals .
Network/Software Support Team	Can access assigned Catalog Tasks and update fulfillment progress.
System Administrator	Has full access to configure, monitor, and maintain the application.

Additional security measures include:

- Role-based access control using **Access Control Lists (ACLs)**.
- Approval actions restricted to the assigned manager.
- Secure handling of request data using ServiceNow's built-in security framework.
- Controlled cross-scope access only where required for notifications and workflow execution.

6. Scalability and Extensibility

The proposed architecture is designed to support future enhancements without requiring major redesign.

Possible future improvements include:

- Adding multi-level approval workflows involving department heads or software asset managers.
- Integrating a Software License Inventory module to perform real-time license availability checks.

- Integrating with software deployment tools such as Microsoft SCCM or Microsoft Intune for automated software installation.
- Reusing the same architecture for additional Service Catalog services such as hardware requests, VPN access, application access, and account provisioning.
- Implementing dashboards and reports for monitoring request trends, approval times, and software usage.

Summary

The proposed solution architecture leverages ServiceNow's native platform capabilities to build a secure, scalable, and maintainable Software Installation Request Management System. The layered architecture separates the user interface, business logic, and data management components, ensuring clear responsibilities and easier maintenance. By combining the **Service Catalog**, **Flow Designer**, **Approval Engine**, **Business Rules**, and standard ServiceNow tables, the solution delivers an efficient end-to-end workflow that improves request management, enhances user experience, and supports future organizational growth.